

From Detection to Attribution: Tracking and Explaining Stream Change in Antarctica from Terrain Alone

NASA ROSES-2025 F.5: Future Investigators in NASA Earth and Space Science and Technology (FINESST)
Division: Earth Science (EARTH25), Antarctic cryosphere / climate Deadline: 14 July 2026 (11:59 PM EDT)
Roles: Grad student is the Future Investigator (FI), lead and main author; advisor is PI on record. Up to ~\$50k/yr for up to 3 years.

Summary. A working tool already finds where Antarctic streams are reshaping the land, but it stops at location: it says where, never why. This project starts at that line. I'll connect the measured change to the processes driving it (heat, melt, thawing ground), make the detector work across different sensors and all four valleys, and attach a real uncertainty value to every pixel. I've already rebuilt and verified the existing pipeline myself and recovered the first driver signal, so this is a concrete next step I'm ready to run, not a proposal resting on untested machinery.

1. The setup and the gap

The McMurdo Dry Valleys (MDVs) are the largest ice-free region in Antarctica, a cold polar desert of bare rock and gravel. For roughly a century they were treated as the most geomorphically stable place on Earth, but that's no longer holding. The system sits right at the energy threshold for melting, so small increases in absorbed heat are enough to melt ice and degrade permafrost. The visible result is ephemeral streams: channels that flow for a few weeks each summer, transport sediment, and reshape the channel before going dry again.

Those streams are the clearest climate signal available in an otherwise stable landscape. If they accelerate, that's an early indicator worth taking seriously.

Barlow (2026) [1] built the tool that measures them, and it rests on three components worth naming:

1. Terrain-only segmentation. A U-Net (an image-segmentation CNN) traces stream channels from the shape of the ground alone (elevation, slope, aspect). There's no water signal to key off, since the channels are dry most of the year.
2. Validating the free satellite data. Airborne lidar is the accuracy benchmark but only exists for 2001 and 2014. Extending past 2014 means using REMA (free satellite elevation models). She showed REMA is good enough via ICP (a 3D alignment method) and NMAD (a median-based, outlier-robust spread estimate that's far more reliable here than standard deviation).
3. Measuring change. Differencing an old elevation surface from a newer one inside the channels gives a DEM-of-Difference (DoD). Only changes above the noise floor ($\text{LOD95} = 1.96 \times \text{NMAD}$) count. Summed per stream, that yields erosion/deposition rates and even acceleration.

The gap. Barlow's work ends at *description*. It maps where change happens (Denton Hills erodes hardest; parts of Taylor Valley are accelerating) but explicitly leaves two things for future work: (a) linking the change to its physical and climate drivers, and (b) making the model work beyond the MDVs. That's the natural space for a Future Investigator project: moving from a tool that *watches* to one that *explains and predicts* (Fig. 1).

From detection (Barlow 2026) → attribution & prediction (this FINESST project)

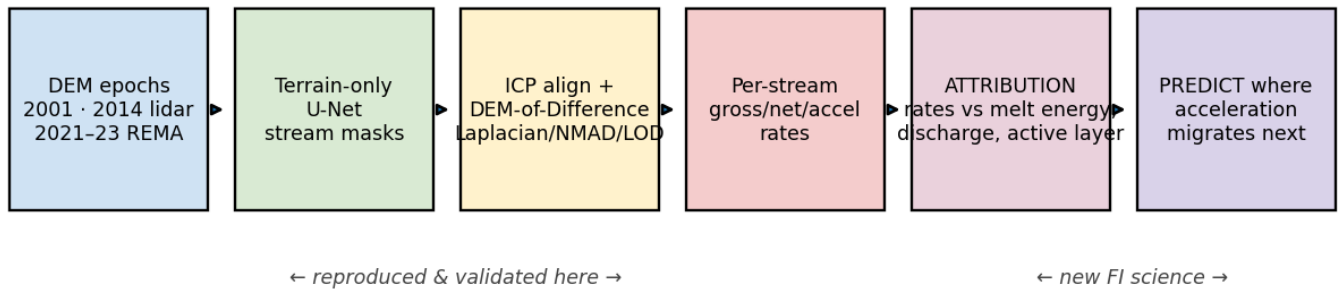


Figure 1. Left (detect): Barlow's pipeline, which I've rebuilt and verified. Right (explain and predict): the new work proposed here.

2. What I'll do (objectives and hypotheses)

#	Objective	Hypothesis	Why it's new
O1	Attribution (the headline). Relate each stream's change rate to its drivers: temperature, insolation, positive-degree-days (PDD) (cumulative above-freezing heat), melt-season discharge, glacier melt, and active-layer (thaw) depth, drawn from the LTER field network plus ERA5/AMPS reanalysis.	H1: Change is governed by <i>cumulative melt energy</i> , not raw water volume. An energy-based model outperforms a discharge-only model and explains the residual scatter in Fig. 5.	Barlow lists this as future work; it hasn't been quantified.
O2	Generalization. Make the terrain-only detector work across both lidar (2001/2014) and REMA, and across all four valleys. The obstacle is domain shift: lidar and satellite represent the surface differently, which degrades the satellite epoch.	H2: A sensor-aware correction (conditioned on slope/aspect, per pixel) closes most of the lidar-vs-REMA gap.	Barlow flags sensor generalization as future work.
O3	Calibrated uncertainty. Replace the single global noise value with a per-pixel, sensor-aware one, so every change map carries a defensible confidence layer.	H3: Conditioning NMAD on slope, aspect, and sensor produces a 95% bound that actually holds 95% of the time.	Barlow uses one global value per epoch pair.

The unifying payoff: a model that takes a driver field and predicts where acceleration migrates next, which is the operationally useful form of the climate-indicator idea.

3. Approach

Data (all free/public, already pulled; see §6). Elevation: MDV lidar 2001 (2 m) and 2014 (1 m), plus REMA satellite (2 m, 2021–23). From each I build the standard terrain layers: elevation, slope, aspect, profile curvature (signed, so it distinguishes concave channels from convex ridges), MFD flow accumulation, and lidar intensity. Drivers: LTER meteorology, 21 stream gauges, glacier melt for 7 glaciers, and continuous soil temperature/moisture as an active-layer proxy; ERA5 and AMPS reanalysis fill spatial gaps. Labels: LTER stream centerlines, a public stand-in for Barlow's private hand-drawn training tiles (the one access

constraint; see §7).

O1, attribution. For each gauged stream and epoch I'll: (i) take the change rate from the DoD inside the channel (already operational, Fig. 2); (ii) assemble a per-stream energy time series (PDD, insolation, discharge, thaw depth); (iii) fit rate-vs-driver models (hierarchical regression and random forest) with leave-one-stream-out validation; and (iv) test H1 directly: energy-based model against discharge-only. The discharge relationship in Fig. 5 is real but leaves structured residuals, which is exactly what an energy model should resolve. For streams with three time points, I'll regress *acceleration* against *driver trends*.

O2, generalization. Quantify the lidar-vs-REMA difference over stable ground as a function of slope and aspect, learn a correction, retrain the U-Net with both sensors represented, and evaluate F1 across all four valleys and both sensors. My prior work (§5) already shows this architecture holding up on an entirely different landscape, so the risk here is bounded.

O3, uncertainty. Replace the global NMAD with one conditioned on (slope, aspect, sensor), output it as a per-pixel layer, and verify calibration by confirming the 95% bound is exceeded about 5% of the time on stable ground. Fig. 4 is the global version of this result.

Reproducibility. The pipeline is fully scripted: re-run from source and the outputs match, with robust statistics and CRS checks enforced throughout. (One catch that validated the QC: an early nodata bug inflated the noise estimate to 2.7 m; it was caught and fixed before any result was trusted.)

4. Preliminary results

The change-detection foundation this project builds on already works. Rebuilding Barlow's pipeline on Taylor Valley reproduces results inside her published ranges (below), and a first driver analysis surfaces the attribution signal that O1 develops.

(a) Change detection across all three epochs (Fig. 2).

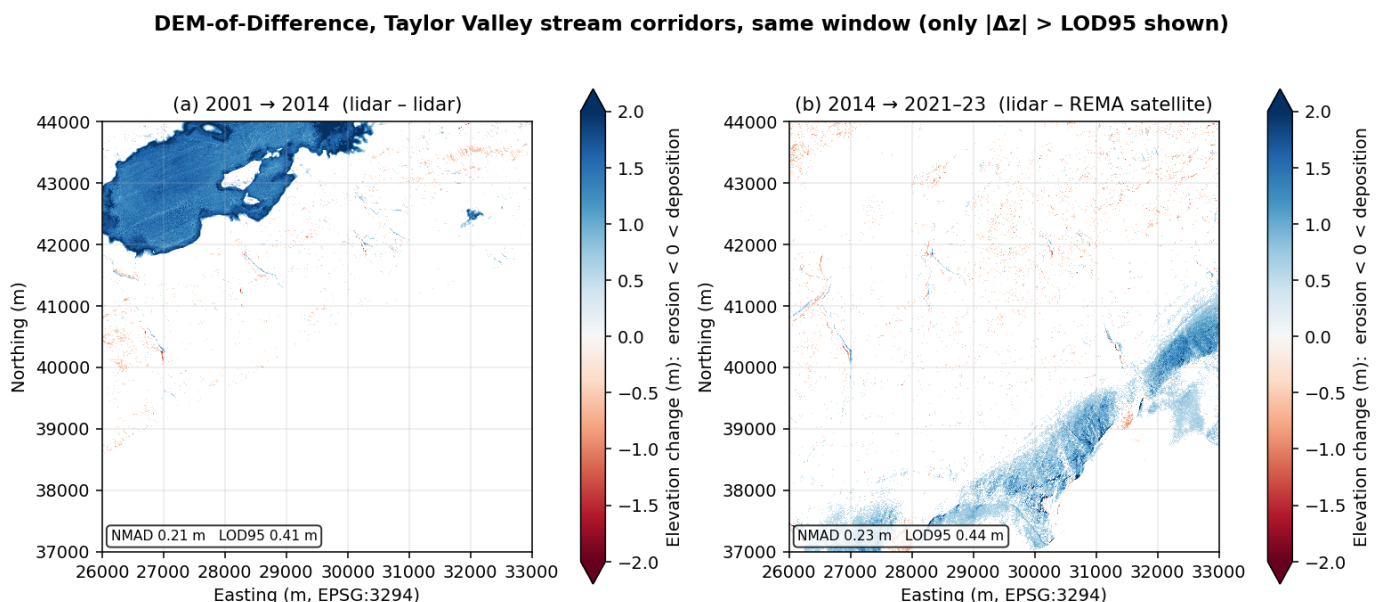


Figure 2. DoD maps over the same Taylor Valley stream-corridor window, showing only changes above the noise floor. Left: 2001→2014 (lidar vs lidar). Right: 2014→2021-23 (lidar vs REMA). Red is erosion, blue is deposition. The big flat patch on the left is Lake Fryxell rising ~1.5 m, real change, but water, so an automated screen detects flat water surfaces and drops them from every stream rate.

Epoch	NMAD (noise)	LOD95 (detection floor)	Barlow's range	Match
2001→2014 (lidar-lidar)	0.21 m	0.41 m	NMAD 0.07-0.46 / LOD95 0.15-0.92	✓
2014→2021-23 (lidar-REMA)	0.23 m	0.44 m	NMAD 0.19-0.53 / LOD95 0.37-1.04	✓

ICP behaved as expected: it improved alignment on terrain with relief and correctly added nothing on the flat valley floor, where there's no 3D structure for it to lock onto.

(b) Per-stream rates (Fig. 3), the direct input for O1.

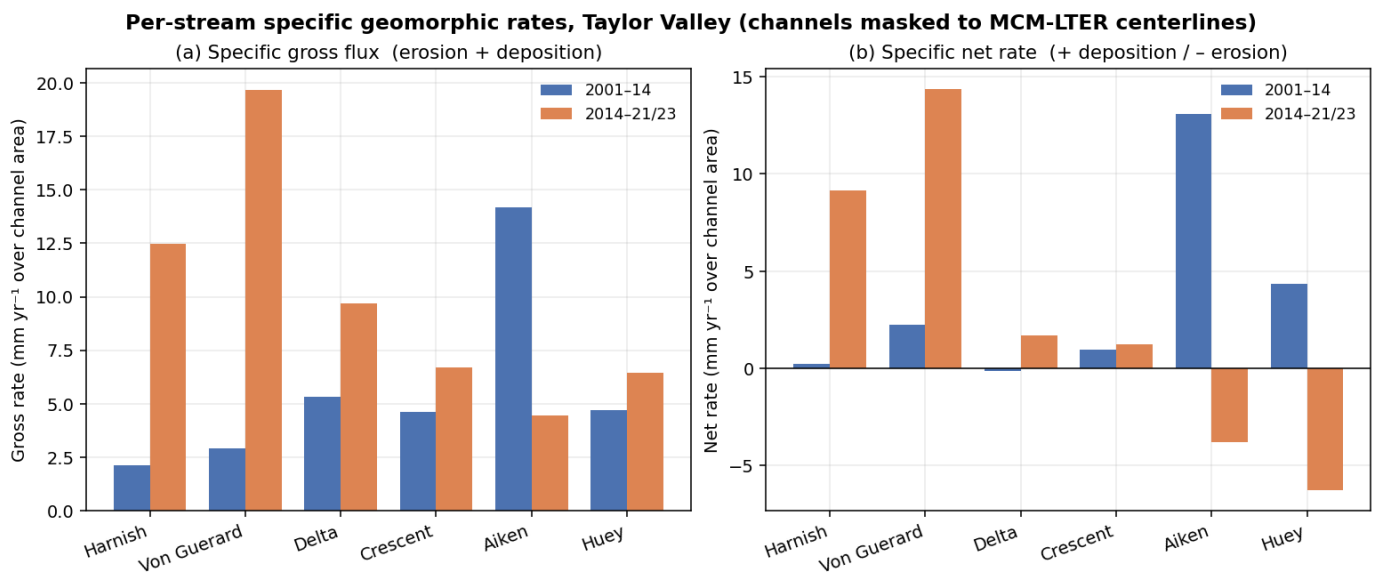


Figure 3. Per-stream rates for six gauged Taylor Valley streams, reported as mm/yr over the channel area rather than raw volume. Gross is total activity (erosion plus deposition); net is the balance (positive = building up, negative = wearing down). Area-normalizing matters because the two epochs don't cover identical footprints (the 2001 lidar swath is narrower), so raw volumes would mix real change with coverage differences.

(c) The noise is Laplacian, not Gaussian (Fig. 4), the basis for O3.

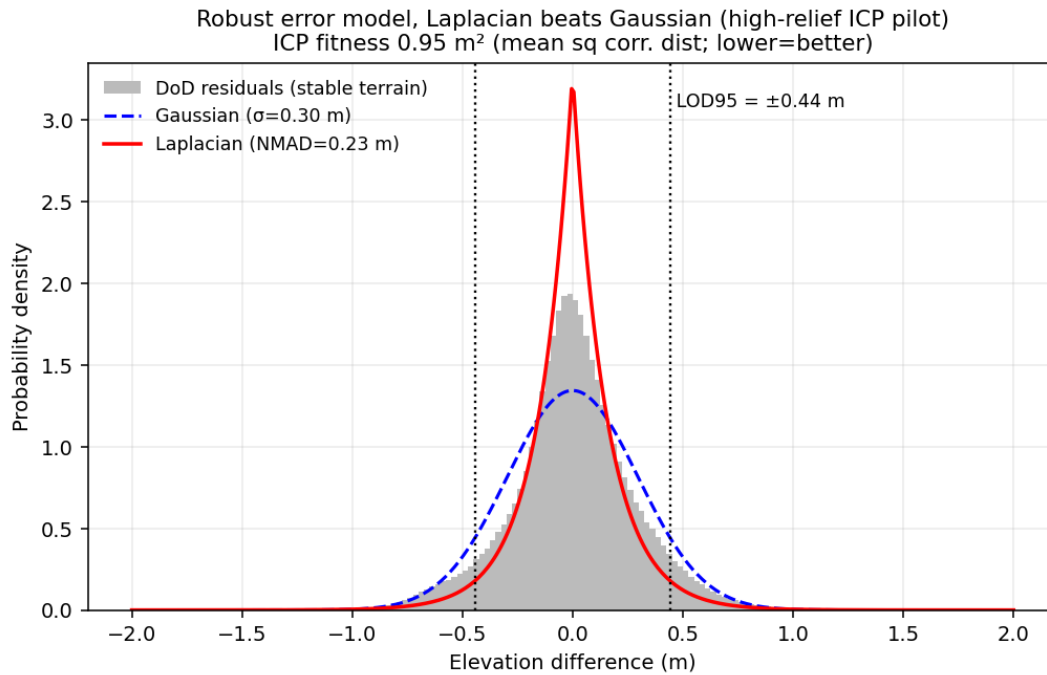


Figure 4. Residuals on stable ground are sharply peaked with heavy tails, i.e. Laplacian rather than the usual bell curve. Assuming a bell curve (and using σ) would let the tails inflate the noise estimate, which is why NMAD is the right choice and the starting point for the per-pixel uncertainty layer in O3.

(d) A first attribution signal (Fig. 5).

Preliminary attribution: sediment flux vs melt discharge, n = 6 gauged streams per epoch (lake-level rise screened)

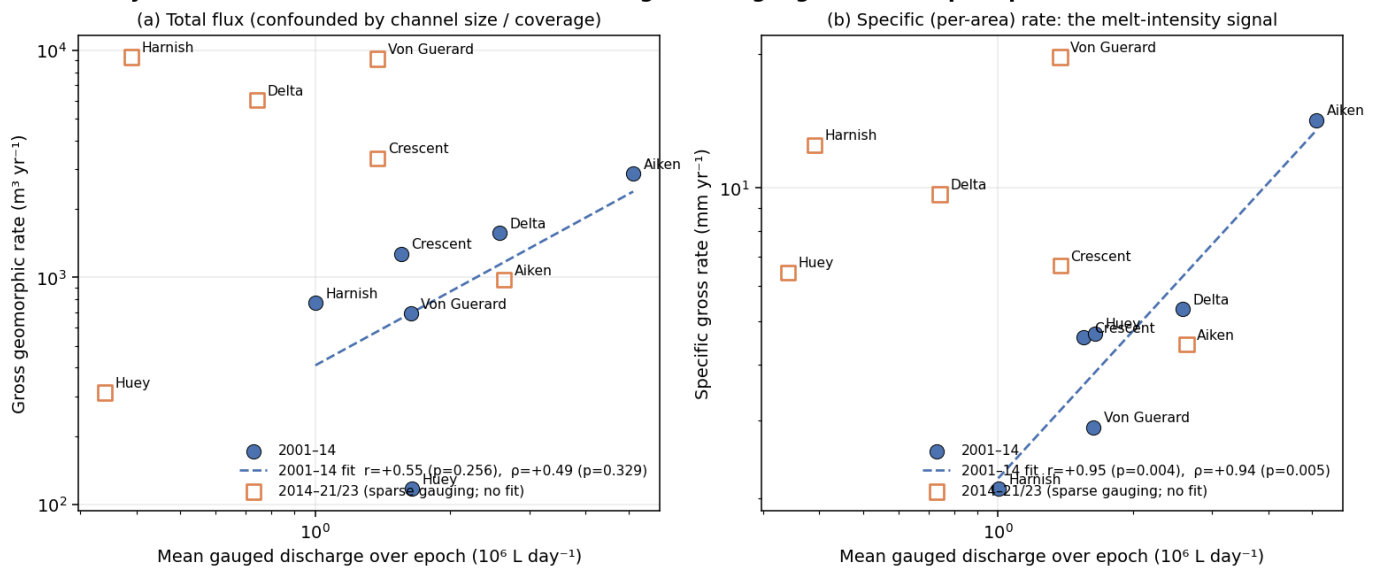


Figure 5. Each stream's rate against its mean gauged melt discharge (lake signal screened out). In the lidar epoch the per-area rate scales strongly with discharge, $r = +0.95$ ($p = 0.004$), Spearman $\rho = +0.94$ ($p = 0.005$), and dropping any single stream doesn't break it. In the satellite epoch there's no coherent relation, because post-2015 gauge records are thin (as few as 48 gauged days for one stream), so no fit is drawn. That contrast is the case for O1: where gauges are dense the melt-sediment link is strong, so the project swaps patchy gauge data for modeled melt energy (PDD / insolation / thaw) that exists everywhere, every year.

5. Why I'm positioned to do this

I've built and run this exact machinery (U-Net segmentation on terrain layers, ICP alignment, DoD change detection) on a very different landscape: detecting channels, roads, and disturbance scars in the Appalachian Plateau from elevation alone. That shows the approach transfers across sensors and biomes, which is precisely O2's risk, and that I can operate the full pipeline end to end.

6. Data: requirements and availability

Every dataset the project needs is free/public, and I've already assembled the full stack (scripted, so it regenerates from source). Data availability isn't a schedule risk. The only access-gated item is Barlow's training labels (addressed in §7).

Category	Source	Status
A. Terrain: MDV lidar 2001 + 2014; REMA 2021-23	free/public	✓ in hand
Per-epoch terrain layers (slope/aspect/curvature/flow/intensity)	scripted from terrain	✓ shown on Taylor pilot
B. Labels: LTER stream centerlines (QC)	EDI knb-lter-mcm.6007	✓ in hand
Barlow's hand-drawn training tiles	author-gated	▲ the one access constraint (see §7)
C. Drivers: LTER meteorology, 21 gauges, glacier melt, soil/thaw	EDI	✓ in hand
Reanalysis: ERA5, AMPS	CDS; GDEX	✓ in hand
D. Cross-valley: all four valleys, lidar + REMA	OpenTopography/PGC	✓ DEMs in hand; per-valley layers to build
E. Validation: high-res imagery; published rates	PGC (restricted); literature	optional / cross-check

7. Plan, timeline, risks, and data sharing

My role and development. I'm the lead and main author; my advisor mentors and is PI of record. I'll present at AGU and a SCAR/ISAES Antarctic venue, publish the O1 attribution result as its own paper, and take coursework in Bayesian/hierarchical modeling and remote-sensing uncertainty.

Timeline (3 years).

Year	Deliverables
Yr 1	Build the per-stream driver time series; first O1 attribution model on Taylor Valley (extending Fig. 5); prototype the per-pixel uncertainty layer (O3). Output: attribution paper drafted.
Yr 2	Correct the lidar↔REMA domain shift and generalize the detector to all four valleys (O2); run attribution basin-wide; AGU talk. Output: O2 paper.
Yr 3	Predict where acceleration migrates next; full three-epoch acceleration attribution; release the uncertainty product (O3). Output: synthesis/prediction paper plus public datasets.

Risks and mitigations. (1) *Barlow's training labels are gated*. I'll request them from her group; if that stalls, the LTER centerlines already serve as a stand-in (Figs. 2–3 were built on them), and my prior work shows I can build labels from scratch. (2) *REMA coverage after 2014 is uneven*. Prioritize Taylor Valley (full three-epoch coverage) for the acceleration work; treat other valleys as two-epoch. (3) *Gauge records have gaps*. That's a core reason O1 relies on melt energy from reanalysis rather than raw gauge discharge alone.

Data sharing. All inputs are free/public and cited. My outputs (change maps, per-stream rate tables, uncertainty layers, stream-boundary polygons) will be released open-access (DOIs via Zenodo/EDI) alongside the code. Heavy regenerable rasters stay out of version control; scripts, figures, and tables are tracked. Everything carries explicit CRS, method, and uncertainty metadata, consistent with NASA's open-science requirements.

References

- [1] Barlow, M. C. (2026). *A Comprehensive Spatial Analysis of Stream Boundary and Geomorphological Change Detection: McMurdo Dry Valleys, Antarctica*. PhD Dissertation, Univ. of Houston (chair: C. L. Glennie). 240 pp.
- [2] Barlow, M. C., Zhu, X., & Glennie, C. L. (2022). Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. *Remote Sensing* 14(1):234. doi:10.3390/rs14010234. (*Dissertation Ch. 4, the proof of concept.*)
- [3] Höhle, J., & Höhle, M. (2009). Accuracy assessment of DEMs by means of robust statistical methods. *ISPRS J. Photogramm. Remote Sens.* 64(4):398–406. (*Origin of NMAD.*)
- [4] Howat, I. M., et al. (2019). The Reference Elevation Model of Antarctica (REMA). *The Cryosphere* 13:665–674.
- [5] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI* 234–241.
- [6] McMurdo Dry Valleys LTER datasets (stream discharge, meteorology, glacier mass balance, soil), Environmental Data Initiative, knb-lter-mcm.*.
- [7] Hersbach, H., et al. (2020). The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* 146:1999–2049.

Plain-language version of finesst_proposal.md: same science and same real numbers, more direct voice. Figures from barlow/build/_finesst_figures.py. Not a submitted proposal.